

Exercises Midterm

- (30%) A plant can manufacture five products P_1, P_2, P_3, P_4 and P_5 . The plant consists of two work areas: the fabrication area A_1 and the assembly area A_2 . The time required to process one unit of product P_j in work area A_i is p_{ij} (in hours), for $i = 1, 2$ and $j = 1, 2, \dots, 5$. The weekly capacity of work area A_i is C_i (in hours). The company can sell all it produces of product P_j at a profit m_j , for $j = 1, 2, \dots, 5$.

The plant manager thought of writing an LP to maximize profit, but never actually did for the following reason. From past experience, he observed that the plant operates best when at most two products are manufactured at a time. He however believes that if he uses linear programming, the optimal solution will consist of producing all five products. Formulate the corresponding LP (define all decision variables properly). With your new expertise in LP, can you uphold, debunk or make no conclusive statement regarding the plant manager's belief of the optimal LP solution? Explain briefly.

- (30%) An intermediate tableau shown below was generated using the Two-Phase method to solve the following LP. Continue the Two-Phase method and generate the next iteration or BFS. Show all your calculations and briefly describe the basis of your calculations.

$$\text{Max } Z = 3x_1 - 6x_2 + x_3$$

subject to

$$x_1 + x_2 + x_3 \geq 8$$

$$2x_1 - x_2 = 5$$

$$-x_1 + 3x_2 + 2x_3 \leq 7$$

$$x_1, x_2, x_3 \geq 0$$

	x_1	x_2	x_3	s_1	R1	R2	s_3	
	0	0	0	0	-1	-1	0	0
x_2	0	1	$2/3$	$-2/3$	$2/3$	$-1/3$	0	$11/3$
x_1	1	0	$1/3$	$-1/3$	$1/3$	$1/3$	0	$13/3$
s_3	0	0	$1/3$	$5/3$	$-5/3$	$4/3$	1	$1/3$

- Consider the following LP where a and b are real numbers.

$$\text{Max } x + y$$

$$a \cdot x + b \cdot y \leq 1$$

$$x, y \geq 0$$

- (10%) For what values of a and b will the LP be infeasible?
 - (10%) Consider the cases when a and b are either positive, negative or zero. What can you conclude about the LP's status for each of these cases?
- (20%) Formulate the dual of the LP in Problem 1.